Personalized Grocery and Diet Planning Application

Course: CP317

Group Number: 3

Roles:

Q/A & Testing: Zulqarnain Ali, Fatima Afzal

Product Owner: Dezirae Asante

Software Engineers: Bassil Obeidi, Toshani Settivari, Dezirae Asante

UI/UX Designers: Zulqarnain Ali, Fatima Afzal

Scrum Master: Bassil Obeidi

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1. Abstract

Keeping up with dietary goals can be difficult when grocery shopping, meal planning, and nutrition tracking all happen separately. People often rely on multiple apps or try to keep track of everything in their heads. This takes time and effort, from looking up foods and checking nutritional information to deciding what to cook and tracking progress. Having to manage all these tasks separately can make it harder to stay organized and consistent.

Our project proposes a Personalized Grocery and Diet Planning Application that brings these tasks together in one place. Users will be able to set dietary goals, track the groceries they have, get meal suggestions based on available ingredients, and automatically calculate nutritional information. The goal is to make everyday food decisions easier and help users make better use of the groceries they buy. We plan to build a working prototype that shows how these features can simplify grocery and diet planning. The final features will depend on the project's scope and what we can realistically implement.

2. Project Description & Objectives

a. Description

We are developing a **Personalized Grocery and Diet Planning Application**, a personalized system that helps users plan meals, manage groceries, and monitor their dietary goals using the groceries they own. The system has three main parts:

1. **User Profile and Goal Setup:**
2. **Grocery Inventory and Meal Planning/Recommendations:**
3. **Meal Logging, Progress Report and Historical Analytics:**

b. Objectives

The project will aim to:

1. **Develop a personalized onboarding process:** Collect goals, preferences, dietary restrictions, and basic information.
2. **Create a user profile from this information:** Used to personalize subsequent meal and dietary recommendations. .
3. **Calculate nutrition targets:** Determine daily calorie and macro (protein, carbs, fat, fibre) targets based on the user's profile and goals.
4. **Track weekly groceries:** Allow users to add, edit, and delete grocery items with their nutrition information.
5. **Personalize meal and grocery planning:** Suggest meals based on user goals and available ingredients, and generate grocery lists when needed.

6. **Track meals and progress:** Log food quantities and automatically calculate calories and macros consumed and remaining.
7. **Support and analyze dietary goals:** Adapt to preferences, allergies, and budget considerations while providing historical nutrition and grocery item trends.

3. Initial Product Backlog

Legend:

- UI: User Interface
- PROF: User profile
- MEAL: Meal planning
- GROC: Grocery Planning
- NUTR: Nutrition Tracking
- REP: Reports/Analytics

User ID	Story Title	Story
PROF01	Personalized Goals	As a weightlifter, I want to spend less time planning what groceries to buy and calculating how to hit my macro targets so that I can spend more time training and doing other things I care about.
PROF02	Dietary Preferences and Restrictions	As a health-conscious person, I want to specify my dietary preferences, allergies and priorities so that the application can tailor its recommendations to me.
MEAL01	Meal Suggestions	As a busy person, I want the application to suggest meals based on my goals and preferences so that I spend less time deciding what to eat.
MEAL02	Meal Suggestions	As a conscientious foody, I want to see meals that can be made from ingredients I already have or plan to purchase so that I can make more efficient use of my groceries.
GROC01	Grocery List	As a meal planner, I want the application to generate a grocery list from my planned meals and/or personalized targets so that I do not have to manually compile one.
NUTR01	Nutrition Tracking	As a user with nutritional goals, I want to track the nutritional content of my meals so that I can see how well my diet aligns with

		those goals.
REP01	Historical Trackings	As a returning user, I want to view historical information about my nutrition and grocery spending so that I can identify trends in my habits over time, and optimize my long-term habits.

4. Ethical Considerations

We will consider the following ethical concerns throughout the design and development of the system:

1. Data Privacy and Security

The application will collect information such as age, weight, dietary preferences, allergies, nutrition data, and potentially grocery spending. This information could be sensitive if improperly accessed or linked to personally identifying information. We will minimize the amount of personal data collected, allow users to skip any or all onboarding questions, and provide a functional default experience using generic settings when personalized information is not provided. Users will be able to provide additional information at their discretion to improve personalization. Critically, we will also implement appropriate authentication and access controls to protect any user data gathered.

2. Inclusivity and Representation

The application's onboarding process and personalization features may fail to accommodate users whose dietary practices, preferences, or circumstances are not represented by predefined options. This could result in certain users being excluded from aspects of the application's functionality or receiving less suitable recommendations, potentially constituting a form of inadvertent discrimination. While it is not feasible to anticipate and accommodate every individual circumstance, we will aim to provide diverse onboarding options, aim to allow users to specify preferences beyond predefined categories, and transparently communicate the limitations of the system's personalization capabilities.

3. Accessibility and Usability

The application should accommodate users with varying abilities and levels of technical experience. An inaccessible or unnecessarily complex interface may prevent certain users from effectively accessing the application's functionality, creating barriers to participation. We will aim to provide intuitive navigation, readable text, accessible input methods, and clear presentation of nutritional information without relying exclusively on colour or visual indicators.

4. Accuracy and User Safety

The application will provide automated nutritional calculations and dietary recommendations that may influence users' food choices and dietary habits. Inaccurate calculations or inappropriate recommendations could lead users to make dietary decisions that are unsuitable for their individual circumstances. We will use established formulas and clearly defined nutritional inputs, while transparently communicating the nature and limitations of the system. Users will be informed that automated recommendations are intended for general informational purposes rather than professional medical or nutritional advice, and that reliance on such recommendations may carry risks, particularly for individuals with specific medical or dietary requirements. Importantly, users will be advised to consult with qualified professionals for specific medical or dietary conditions.

5. Team Blog

(attached alongside this document)